

EXPERIMENT NO 8

8.1.1 Multiplication of two square matrices

Algorithm

Step 1: Start the program.

Step 2: Input the dimension n of the square matrices.

Step 3: Input the elements of the first matrix A of size $n \times n$.

Step 4: Input the elements of the second matrix B of size $n \times n$.

Step 5: Initialize $i = 0$.

Step 6: Check if $i < n$.

- If **No**, go to **Step 18**.
- If **Yes**, go to **Step 7**.

Step 7: Initialize $j = 0$.

Step 8: Check if $j < n$.

- If **No**, go to **Step 16**.
- If **Yes**, go to **Step 9**.

Step 9: Set $C[i][j] = 0$.

Step 10: Initialize $k = 0$.

Step 11: Check if $k < n$.

- If **No**, go to **Step 14**.
- If **Yes**, go to **Step 12**.

Step 12: Compute

$$C[i][j] = C[i][j] + A[i][k] \times B[k][j].$$

Step 13: Increment $k = k + 1$ and go back to **Step 11**.

Step 14: Increment $j = j + 1$.

Step 15: Go back to **Step 8**.

Step 16: Increment $i = i + 1$.

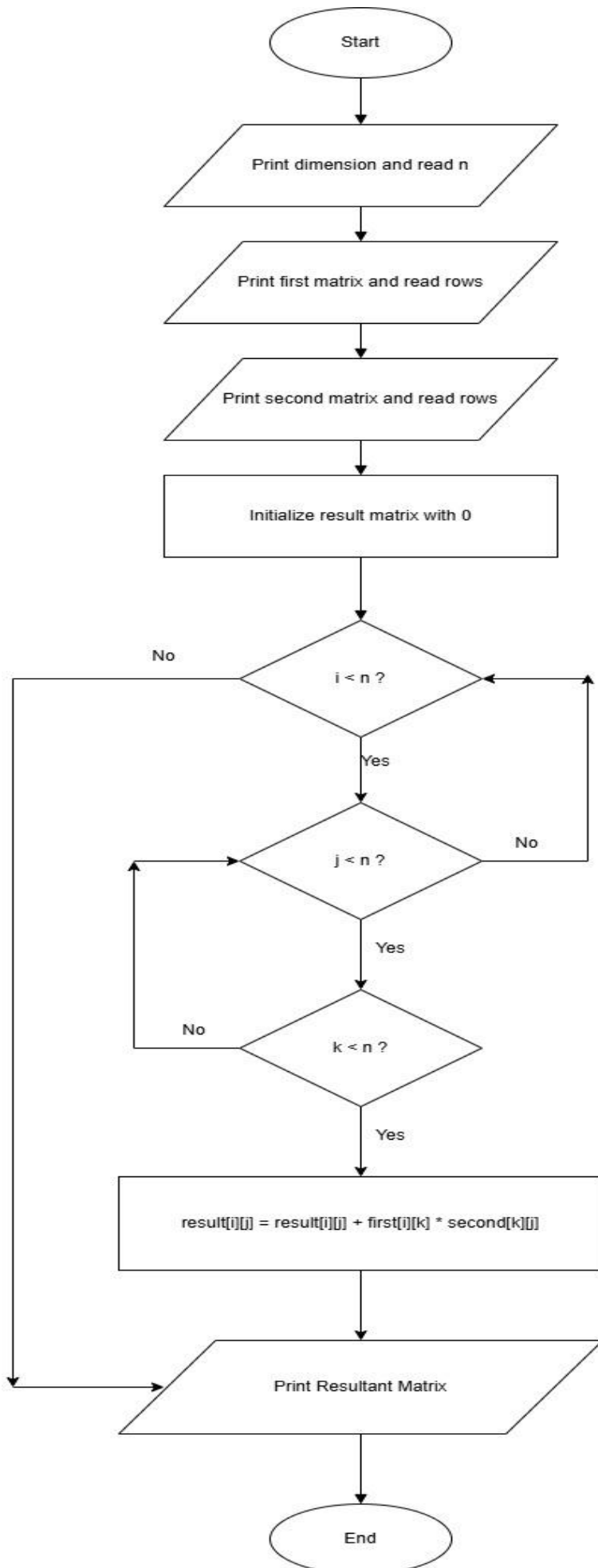
Step 17: Go back to **Step 6**.

Step 18: Print the resultant matrix C .

Step 19: End the program.

Flowchart

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PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    row = list(map(int, input().split()))
    A.append(row)
print("second matrix:")
B = []
for i in range(n):
    row = list(map(int, input().split()))
    B.append(row)
result = []
for i in range(n):
    result.append([0] * n)
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]

print("Resultant Matrix:")
for row in result:
    print(*row)
```

EXCECUTION

EXPERIMENT NO 8

8.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input.

Input Format:

- The first line prints the "dimension: " and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the resultant matrix.

Sample Test Cases

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